

Growth rate measurements

Mean annual linear extension rates (upward linear growth), expressed in $\text{cm}\cdot\text{yr}^{-1}$ (Table S1), were determined for each coral core by measuring the distance between successive low-density bands over the last six years of growth (2010–2016), excluding the tissue layer (i.e., the last two years of growth; Fig. S1). The uncertainty of the linear extension rates was calculated from two independent sets of measurements (Canesi et al., 2023). The 5-year sampling for DNA analysis was conducted based on the respective average growth rates of the colonies, continuously and starting from the top of the core.

Table S1. Averaged growth rates of the coral colonies (published in Canesi et al. 2023).

Core number	Coral species	Location	Mean linear extension rate ($\text{cm}\cdot\text{yr}^{-1}$)	Figure number of X-Radiograph
I23S2D	<i>Diploastrea heliophora</i>	Papua New Guinea 9°21' S, 152°02' E	0.5	Figure S1-A
I23S2P	<i>Porites lobata</i>	Papua New Guinea 9°21' S, 152°02' E	1.0	Figure S1-B
I26S1D	<i>Diploastrea heliophora</i>	North Palau 7°47' N, 134°35' E	0.5	Figure S1-C
I26S1P	<i>Porites lobata</i>	North Palau 7°47' N, 134°35' E	1.0	Figure S1-D

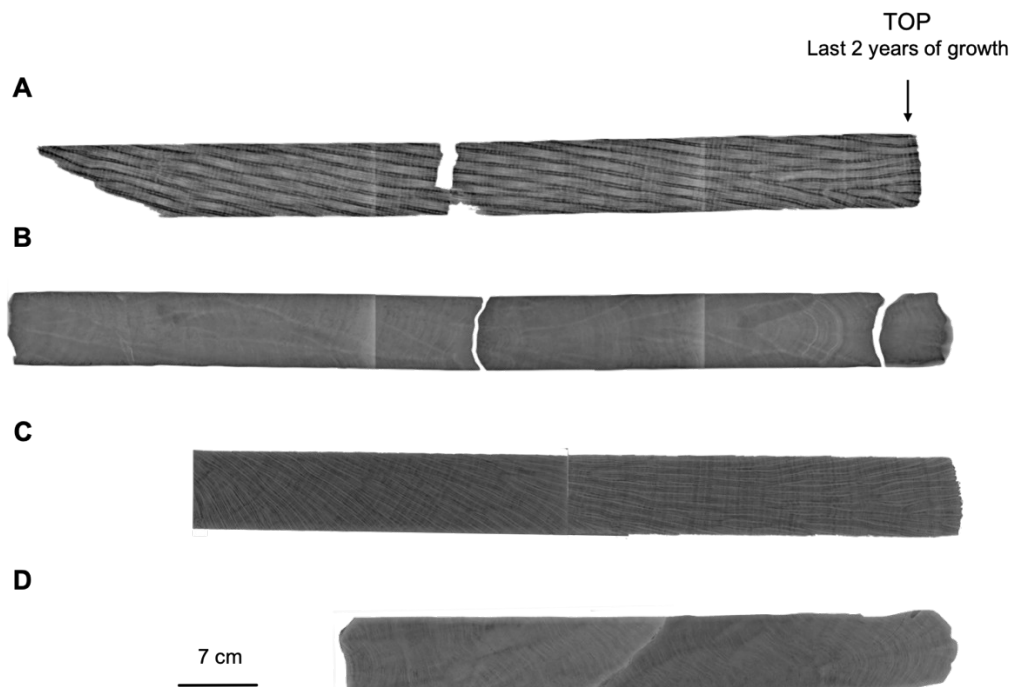


Figure S1. X-ray radiographs of *Porites lobata* (B, D) and *Diploastrea heliophora* (A, C) colonies collected in Papua-New-Guinea (A,B) and Palau (C,D), showing the annual density banding pattern.

Reference

Canesi, M., Douville, E., Montagna, P. *et al.* Differences in carbonate chemistry up-regulation of long-lived reef-building corals. *Sci Rep* **13**, 11589 (2023). <https://doi.org/10.1038/s41598-023-37598-9>